Phase 1: Data Lifecycle Design

1. Objective

The objective of this phase is to define how data is created, stored, used, archived and eventually deleted. A documented lifecycle helps ensure that data is accessible when needed, protected against unauthorized access and retained for the required period.

The project brief specifies three data classes: **Clinical, Billing and Marketing Consent**.

Lifecycle maps

2.1 Clinical data lifecycle

Examples: Patient registration details, consultation records, diagnoses, laboratory results and treatment records.

Lifecycle stage	Description
Create	Clinical data is created when patients register, consult clinicians, undergo laboratory tests or receive treatment.
Store	Records are stored in authorized clinical systems or approved paper-record storage. Access is restricted to authorized personnel.
Use	Authorized healthcare professionals use the records to provide care. Approved, appropriately protected data may also support clinical reporting.
Archive	Records are archived after 7 years under the project brief's lifecycle policy. Archived records remain protected and accessible to authorized personnel when required.
Delete	Records reach the end of the 10-year retention period and are securely purged when legally and operationally permitted. Any applicable preservation requirements must be checked before deletion.

Data owner: Head of Clinical Services
Retention period: 10 years
Archiving point: After 7 years
Protection requirements: Access control, confidentiality, audit trails and secure disposal.

2.2 Billing data lifecycle

Examples: Patient invoices, service charges, payment records and billing adjustments.

Lifecycle stage	Description
Create	Billing records are generated when a patient receives a chargeable service or makes a payment.
Store	Records are stored in an authorized billing or accounting system with appropriate access restrictions.
Use	Authorized finance staff use the records for payment reconciliation, financial reporting and revenue analysis.
Archive	Records are archived after 5 years under the project brief's policy. Archived billing information remains protected.
Delete	Records reach the end of the 7-year retention period and are securely disposed of when applicable legal and financial obligations permit.

Data owner: Finance Manager
Retention period: 7 years
Archiving point: After 5 years
Protection requirements: Restricted access, auditability and preservation of billing history.

The MediLink brief also requires billing records to be immutable. Corrections should be made through adjustment records rather than changing existing billing rows.

3.3 Marketing consent data lifecycle

Examples: Records showing whether an individual has consented to receive promotional communications.

Lifecycle stage	Description
Create	A consent record is created when an individual provides a marketing preference or changes an existing preference.
Store	Consent status, collection date and relevant supporting information are stored in an authorized system.
Use	Authorized personnel use the record to determine whether marketing communications are permitted.
Archive	Records are archived after 1 year under the project brief's policy. Archived records remain access-controlled.
Delete	Records reach the end of the 2-year retention period and are securely deleted when permitted. Any evidence that must be retained to demonstrate compliance should be handled according to an approved policy.

Data owner: Data Steward
Retention period: 2 years
Archiving point: After 1 year
Protection requirements: Restricted access, appropriate use and respect for recorded consent preferences.